MERN STACK

TRAINING TR-103 REPORT DAY 12

8 July 2026

Overview:

The twelfth day of the MERN Stack training focused on JavaScript events and form handling. We learned about the `onclick` event, which allows JavaScript functions to execute when users interact with webpage elements. I also explored the `.value` property to read and update values from form elements. These concepts were applied by creating a simple calculator, helping us understand how JavaScript interacts with HTML to build interactive web applications.

Learning Objectives:

- Understand the `onclick` event in JavaScript.
- Learn how event handlers respond to user actions.
- Understand the use of the `.value` property.
- Learn how to access and modify form input values.
- Build a simple calculator using JavaScript.

JAVASCRIPT

onclick Event

The onclick event is an event handler that executes a JavaScript function when a user clicks an HTML element such as a button, image, or link. It is commonly used to perform actions like displaying messages, performing calculations, or updating webpage content.

Features:

- Executes a function when an element is clicked.
- Used with buttons, images, links, and other HTML elements.
- Makes webpages interactive.

Syntax:

```
< button onclick="change()" > Click Me </button>
```

.value Property

The .value property is used to get or set the value of form elements such as <input>, <textarea>, and <select>. It is commonly used to read user input or update field values dynamically.

Features:

- Reads the value entered by the user.
- Updates the value of form elements.
- Commonly used with forms and user input.

Syntax:

```
document.getElementById("ID-1").value;
```

Calculator (Task)

A calculator was created using HTML, CSS, and JavaScript to perform basic arithmetic operations. The calculator accepts user input, reads values using the .value property, and performs calculations when buttons are clicked using the onclick event.

Features:

- Accepts numeric input from users.

- Performs basic arithmetic operations.
- Uses the onclick event to execute functions.
- Uses the .value property to read and display values.

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .cal {
            justify-content: center;
            align-items: center;
            margin-left: 700px;
            padding: 0.5px;

        }
    </style>
</head>

<body>
    <h1 align="center">TASK </h1>
    <h2 align="center">CALCULATOR</h2>
    <div class="cal">
        <div>
            <input type="text" id="num">
        </div>
        <div class="up">
            <button onclick="addvalue(7)">7</button>
            <button onclick="addvalue(8)">8</button>
            <button onclick="addvalue(9)">9</button>
            <button onclick="addvalue('/')">/</button>
        </div>
        <div class="up">
            <button onclick="addvalue(4)">4</button>
            <button onclick="addvalue(5)">5</button>
            <button onclick="addvalue(6)">6</button>
            <button onclick="addvalue('*')">*</button>
        </div>
        <div class="up">
            <button onclick="addvalue(1)">1</button>
            <button onclick="addvalue(2)">2</button>
            <button onclick="addvalue(3)">3</button>
            <button onclick="addvalue('-')">-</button>
        </div>
        <div class="up">
            <button onclick="addvalue(0)">0</button>
            <button onclick="cleard()">C</button>
            <button onclick="eva()">=</button>
            <button onclick="addvalue('+')">+</button>
        </div>

    </div>
    <script>
        let op = "";

        function addvalue(a) {
            let p = document.getElementById("num");
            p.value = p.value + a;
        }

        function cleard() {
            let p = document.getElementById("num");
            p.value = "";
        }

        function eva() {
            let p = document.getElementById("num");
            let val = p.value;
            let n1 = "";
            let n2 = "";
            for (let i = 0; i < val.length; i++) {
                if (val[i] == "+" || val[i] == "-" || val[i] == "*" || val[i] == "/") {
                    op = val[i];
                    n1 = val.substring(0, i);
                    n2 = val.substring(i + 1);
                    break;
                }
            }

            let ans;

            switch (op) {
                case "+":
                    ans = Number(n1) + Number(n2);
                    break;

                case "-":
                    ans = Number(n1) - Number(n2);
                    break;

                case "*":
                    ans = Number(n1) * Number(n2);
                    break;

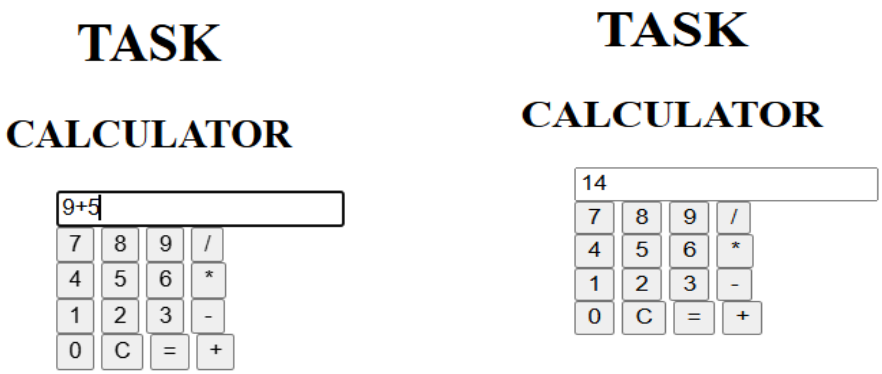
                case "/":
                    ans = Number(n1) / Number(n2);
                    break;

                default:
                    ans = "Invalid";
            }

            p.value = ans;
        }
    </script>
</body>

</html>
```

Output :



Conclusion:

By the end of Day 12, I learned how JavaScript responds to user interactions using the **onclick** event and how the **.value** property is used to read and update form data. I also applied these concepts by developing a simple calculator, strengthening my understanding of interactive web development.